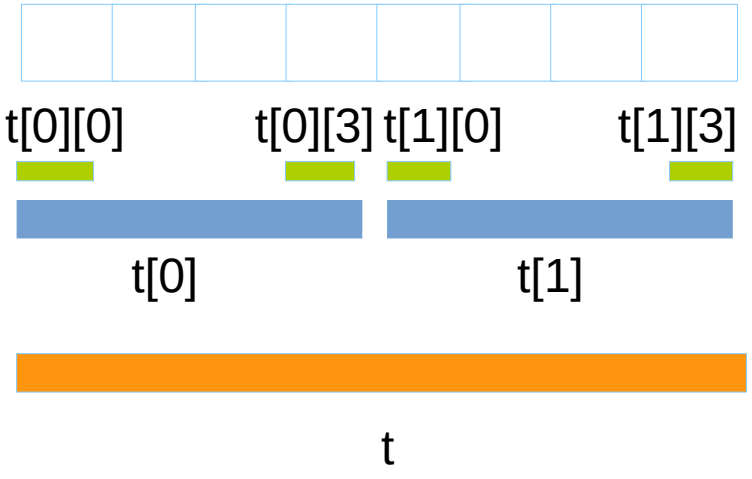
# Move semantics

- Pointers and arrays
- References
- Value vs move semantics
- Right value references
- Move semantics in C++11
- Perfect forwarding
- Traps and pitfalls

# Arrays

- An array is a strictly continuous memory area.
- Arrays do not know their size, but: sizeof(t) / sizeof(t[0])
- Array names could be converted to pointer value to the first element.
- No multidimensional arrays. But there are arrays of arrays.
- No operations on arrays, only on array elements.




```
int t[2][4];

assert(sizeof(t) == 8*sizeof(int));
assert(sizeof(t[0] == 4*sizeof(int));
assert(sizeof(t[0][0])==sizeof(int));

t[0][1] = t[1][1];
// t[0] = t[1];  syntax error
```

# Array decay

- Array to pointer conversion (the first of the “standard conversions”)
- Reference binding can avoid decay

```
void  aDecay( int *ap)      {std::cout << sizeof(ap) << "\n";}
void  pDecay( int (*p)[6]) {std::cout << sizeof(p)  << "\n";}
void noDecay( int (&a)[6]) {std::cout << sizeof(a)  << "\n";}
template <typename T, int N>
void tNoDecay(T (&a)[N]) { std::cout << sizeof(T) << " " << N
                                << " " <<sizeof(a) << "\n"; }
void f()
{
  int t[6] = { 0,1,2,3,4,5 };
  std::cout << sizeof(t) << "\n";  // 24
  aDecay(t);                       //  8
  pDecay(&t);                      //  8
  noDecay(t);                      // 24
  tNoDecay(t);                     // 4 6 24
}
```

# Pointers

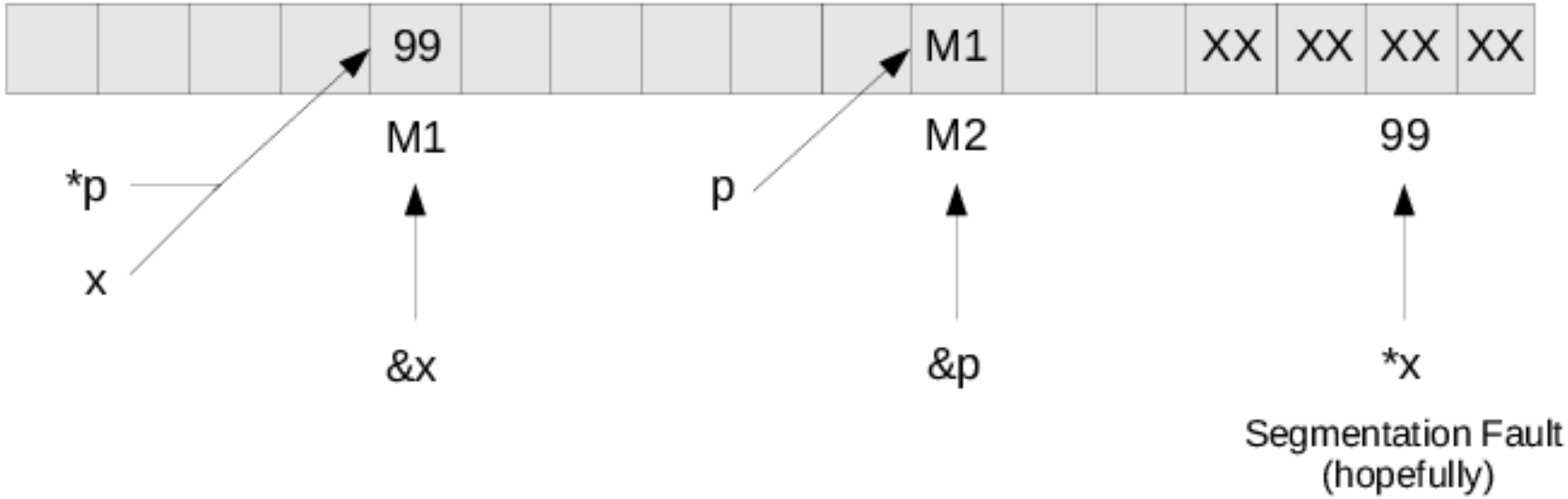
**Pointers in C**

**C Code**

```
int x;
int *p;

x = 99;    //holds a value
p = &x;    //holds an address of a value
```

**Memory**



# Pointers

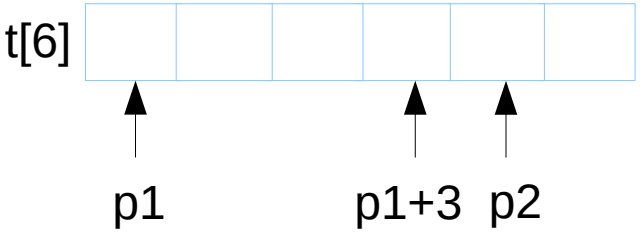
- Pointer is a value that refers to a(n abstract) memory location
- Pointers can refer to ANY valid memory locations (unlike e.g. PASCAL)
- **nullptr** (NULL) is a universal null-pointer value
- Non-null pointers are considered as TRUE value

```
char *ptr = (char *) malloc(1024);

if ( ptr )
{
   // ptr != NULL here
}
```

# Pointer arithmetics

- Integers can be added and subtracted from pointers
- Pointers pointing to the same array can be subtracted
- Pointer arithmetics depends on the pointed type!
- No pointer arithmetics on void *




```
int t[6];
int *p1 = &t[0];
int *p2 = &t[4];

assert( &t[3] == p1+3 );
assert( p2 - p1 == 4 );
assert( p1 + 4 == p2 );
```

```
  p = &t[0];
  p = t;
  p + k == &t[k]
*(p + k) == t[k]
```

# Pointers and arrays

- Pointers and array names can be used similarly in expressions

```
int t[10];
int i = 3;

int *p = t;       // t is used as pointer to the first element

p[i] = *(t + i); // p used as array, t is used as pointer
```

But pointers ARE NOT EQUIVALENT TO arrays !!!

# Pointers and arrays

```c
// s1.c
#include <stdio.h>

int t[] = {1,2,3};

void f( int *par)
{
  printf("%d",par[1]);
  printf("%d",t[1]);
}
int main()
{
  int *p = t;
  printf("%d",t[1]);
  printf("%d",p[1]);
  f(t);
  g(t);
}
```

```c
// s2.c
#include <stdio.h>

extern int *t;

void g( int *par)
{
  printf("%d",par[1]);
  printf("%d",t[1]);
  // the program crashes here
}
```

# Pointers and arrays

```c
// s1.c
#include <stdio.h>

int t[] = {1,2,3};

void f( int *par)
{
  printf("%d",par[1]);
  printf("%d",t[1]);
}
int main()
{
  int *p = t;
  printf("%d",t[1]);
  printf("%d",p[1]);
  f(t);
  g(t);
}
```

```c
// s2.c
#include <stdio.h>

extern int *t;

void g( int *par)
{
  printf("%d",par[1]);
  printf("%d",t[1]);
  // the program crashes here
}
```

# Pointers and arrays

COMPILER EXPLORER

Add... More

Support diversity in C++ with #include <C++>

C++ source #1

C++

```cpp
int t[] = { 1, 2, 3};

int main()
{
  int *p = t;
  int x = t[1];
  int y = p[1];
}
```

x86-64 gcc 9.2 (Editor #1, Compiler #1) C++

x86-64 gcc 9.2

Compiler options...

11010 ./a.out .LX0: lib.f: .text // \s+ Inte

```asm
t:
        .long   1
        .long   2
        .long   3
main:
        push    rbp
        mov     rbp, rsp
        mov     QWORD PTR [rbp-8], OFFSET FLAT:t
        mov     eax, DWORD PTR t[rip+4]
        mov     DWORD PTR [rbp-12], eax
        mov     rax, QWORD PTR [rbp-8]
        mov     eax, DWORD PTR [rax+4]
        mov     DWORD PTR [rbp-16], eax
        mov     eax, 0
        pop     rbp
        ret
```

# Member pointers

- Data Member pointer: Referencing to an offset inside a class
- Member function pointer: Referencing to a (possible virtual) member function of a class
- Works with 2 components: **this + mptr**

```
Type Class::*dmptr;
Type (Class::*fmptr)(P1 par1, P2 par2, …);

Class obj;
Class *ptr = &obj;

obj.*dmptr = …;
ptr->*dmptr = …;

(obj.*fmptr)(par1,par2);
(ptr->*fmptr)(par1,par2);
```

# Member pointers

```
#include <iostream>

class Date
{
public:
  void set (int y, int m, int d);
  int  getYear() const  { return _year; }
  int  getMonth() const { return _month; }
  int  getDay() const   { return _day; }

  void print(std::ostream& os) const;
  void hu();
  void us();
private:
  int _year;
  int _month;
  int _day;

  int Date::*p1;
  int Date::*p2;
  int Date::*p3;
  char sep;
};
```

# Member pointers

```
void Date::hu()
{
  sep = '.';
  p1 = &Date::_year;
  p2 = &Date::_month;
  p3 = &Date::_day;
}
void Date::us()
{
  sep = '/';
  p1 = &Date::_month;
  p2 = &Date::_day;
  p3 = &Date::_year;
}
int main()
{
  Date d;
  d.set(2017,4,20);
  d.hu();
  std::cout << d << std::endl;
  d.us();
  std::cout << d << std::endl;
}
```

```
void Date::set(int y, int m, int d)
{
  _year = y;
  _month = m;
  _day = d;
}

void Date::print(std::ostream& os) const
{
  os << this->*p1 << sep << this->*p2
     << sep << this->*p3;
}

std::ostream& operator<<(
         std::ostream& os, const Date& d)
{
  d.print(os);
  return os;
}
```

```
2017.4.20
4/20/2017
```

# Member pointers

```
  int (Date::*g1)() const;
  int (Date::*g2)() const;
  int (Date::*g3)() const;
};

void Date::hu()
{
  sep = '.';
  g1 = &Date::getYear;
  g2 = &Date::getMonth;
  g3 = &Date::getDay;
}
void Date::us()
{
  sep = '/';
  g1 = &Date::getYear;
  g2 = &Date::getMonth;
  g3 = &Date::getDay;
}
int main()
{
  Date d;
  d.set(2017,4,20);
  d.hu();
  std::cout << d << std::endl;
  d.us();
  std::cout << d << std::endl;
}
```

```
void Date::set(int y, int m, int d)
{
  _year = y;
  _month = m;
  _day = d;
}

void Date::print(std::ostream& os) const
{
  os << (this->*g1)() << sep
     << (this->*g2)() << sep
     << (this->*p3)();
}

std::ostream& operator<<(
         std::ostream& os, const Date& d)
{
  d.print(os);
  return os;
}
```

```
2017.4.20
4/20/2017
```

# Nullptr

- nullptr is a new literal since C++11 of type std::nullptr_t
- Helps to overload between pointers and integer
- Automatic conversion from null pointer of any type and from NULL

```
void f(int*); // 1
void f(int);  // 2

f(0);         // calls 2
f(nullptr);   // calls 1
```

# Reference

- In modern languages definitions hide two important but orthogonal concepts:
  - Allocate memory for a variable
  - Bind a name with special scope to it
- In most languages this is inseparable
- In C++ we can separate the two steps

```
void f()
{
  int i;                  // allocate memory, bind i as name
  int &i_ref = i;         // binds a new name
  int *iptr = new int;    // allocate memory, no binded name
  int &k = *iptr;         // binds a new name to unnamed int area
  delete iptr;            // memory invalidated name k still lives
  k = 5;                  // compiles, later run-time error
}                         // k goes out of scope
```

# Pointer vs reference

- Pointers have extremal element: nullptr
  - nullptr means: pointing to no valid memory
- References always should refer to valid memory
  - Use exception, if something fails

```
if ( Derived *dp = dynamic_cast<Derived*>(bp) )
{
    // use dp as Derived*
}

try
{
  Derived &dr = dynamic_cast<Derived&>(*bp); // may throw
  // use dr as Derived&
}
catch(bad_cast &e) { . . . }
```

# Parameter passing

- Parameter passing in C++ follows initialization semantics
  - Value initialization copies the object
  - Reference initialization just set up an alias

```
void f1( int x, int y) { ... }
void f2( int &xr, int &yr) { ... }

int i = 5, j = 6;

f1( i, j);   int x = i;  // creates local x and copies i to x
             int y = j;  // creates local y and copies j to y

f2( i, j);   int &xr = i; // binds xr name to i outside
             int &yr = j; // binds yr name to j outside
```

# Swap before move semantics

```
void swap( int &x, int &y)
{
  int tmp = x;
  x = y;
  y = tmp;
}

int main()
{
  int i = 5;
  int j = 6;

  swap( i, j);
  assert(i==6 && j==5);
}
```

# Reference binding

- Non-const (left) reference binds only to left value
- Const reference binds to right values too

```
int i = 5, j = 6; double d = 7.0;

swap(i,j);      // ok
swap(i,7);      // error: could not bind reference to 7
int &ir1 = 7;   // error: could not bind reference to 7

swap(i,d);              // error: int(d) creates non-left value
int &ir2 = int(3.14); // error: int(3.14) creates non-left value

const int &ir3 = 7;          // ok, lifetime extension
const int &ir4 = int(3.14); // ok, lifetime extension
```

# Returning with reference

- By default C++ functions return with copy
- Returning reference just binds the function result to an object

```
int f1()
{
  int i = 5;
  return i;   // ok, copies i before evaporating
}

int &f2()
{
  int i = 5;
  return i;   // oops, binds to evaporating i
}
```

# Usage example

```
class date
{
public:
  date& setYear(int y)  { _year = y;  return *this; }
  date& setMonth(int m) { _month = m; return *this; }
  date& setDay(int d)   { _day = d;   return *this; }

  date& operator++() { ++_day; return *this; }
  date  operator++(int) // should return temporary
                { date curr(*this); ++_day; return curr; }
private:
  int _year;
  int _month;
  int _day;
};

void f()
{
  date d;
  ++d.setYear(2011).setMonth(11).setDay(11); // still left value
}
```

# Usage example

```
template <typename T>
class matrix
{
public:
      T& operator()(int i, int j)       { return v[i*cols+j]; }
const T& operator()(int i, int j) const { return v[i*cols+j]; }
    matrix& operator+=( const matrix& other)
    {
        for (int i = 0; i < cols*rows; ++i)
            v[i] += other.v[i];
        return *this;
    }
private:
    // ...
    T* v;
};
template <typename T> matrix<T>  // returns value
operator+(const matrix<T>& left, const matrix<T>& right)
{
    matrix<T> result(left);  // copy constructor
    result += right;
    return result;
}
```

# Left vs right value

- Assignment in earlier languages work the following way:
  <variable> = <expression>, like x = a+5;

- In C/C++ however it can be:
  <expression> = <expression>, like  *++ptr = *++qtr;

- But not all expressions are valid, like a+5 = x;

  An **lvalue** is an expression that refers to a memory location and allows us to take the address of that memory location via the & operator. An **rvalue** is an expression that is not an lvalue

- A rigorous definition of lvalue and rvalue:
  https://accu.org/journals/overload/12/61/kilpelainen_227/

# Value categories

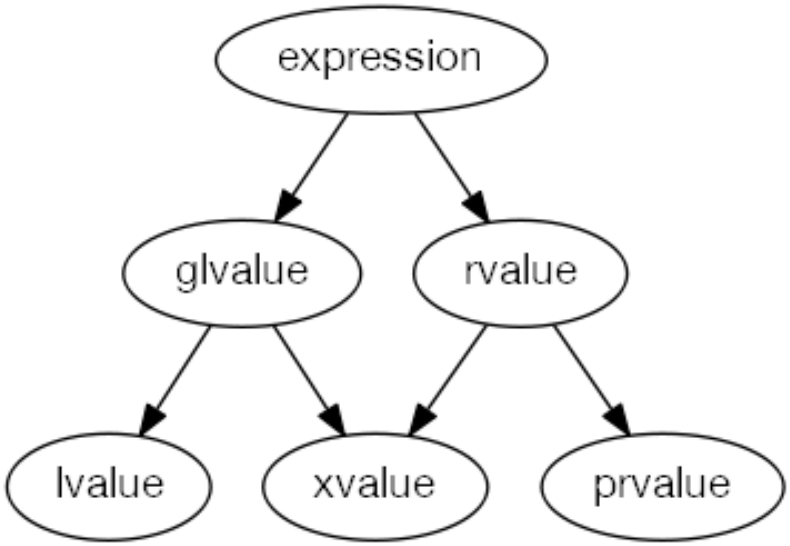
— An lvalue (so called, historically, because lvalues could appear on the left-hand side of an assignment expression) designates a function or an object. [ Example: If E is an expression of pointer type, then *E is an lvalue expression referring to the object or function to which E points. As another example, the result of calling a function whose return type is an lvalue reference is an lvalue. —end example ]

— An xvalue (an “eXpiring” value) also refers to an object, usually near the end of its lifetime (so that its resources may be moved, for example). An xvalue is the result of certain kinds of expressions involving rvalue references (8.3.2). [ Example: The result of calling a function whose return type is an rvalue reference is an xvalue. —end example ]

— A glvalue (“generalized” lvalue) is an lvalue or an xvalue.

— An rvalue (so called, historically, because rvalues could appear on the right-hand side of an assignment expressions) is an xvalue, a temporary object (12.2) or subobject thereof, or a value that is not associated with an object.

— A prvalue (“pure” rvalue) is an rvalue that is not an xvalue. [ Example: The result of calling a function whose return type is not a reference is a prvalue. The value of a literal such as 12, 7.3e5, or true is also a prvalue. —end example ]



# Left value vs. right value

```
int  i = 42;
int &j = i;
int *p = &i;

 i = 99;
 j = 88;
*p = 77;

int *fp() { return &i; } // returns pointer to i: lvalue
int &fr() { return i; }  // returns reference to i: lvalue

*fp() = 66;  // i = 66
 fr() = 55;  // i = 55

// rvalues:

int f() { int k = i; return k; } // returns rvalue

  i = f();  // ok
  p = &f(); // bad: can't take address of rvalue
f() = i;    // bad: can't use rvalue on left-hand-side
```

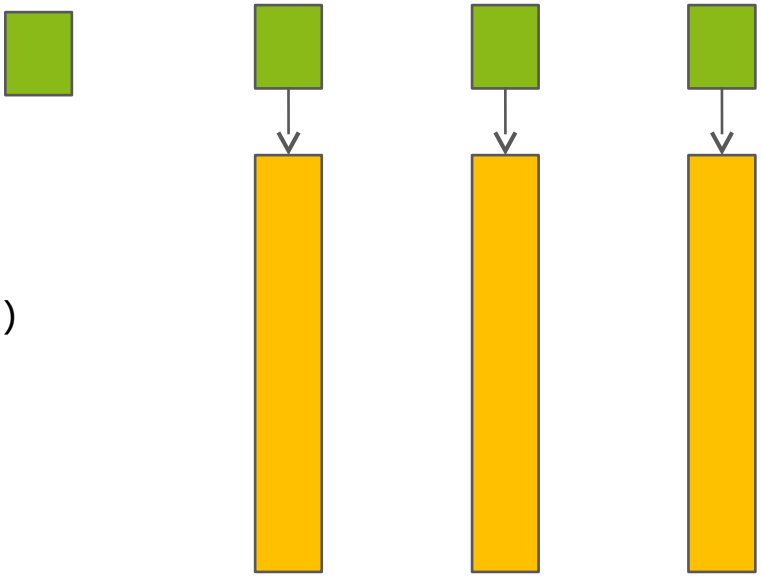
# Value semantics

- C++ has value semantics
  - Clear separation of memory areas
  - Significant performance loss when copying large objects
  - This can lead to improper use of (smart) pointers

# Value semantics

```
class Array
{
public:
   Array (const Array&);
   Array& operator=(const Array&);
   ~ Array ();
private:
   double  *val;
};
Array operator+(const Array& left,const Array& right)
{
   Array res = left;
   res += right;
   return res;
}

void f()
{
   Array b, c, d;
   …
   Array a = b + c + d;
}
```



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public:
   Array (const Array&);
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};
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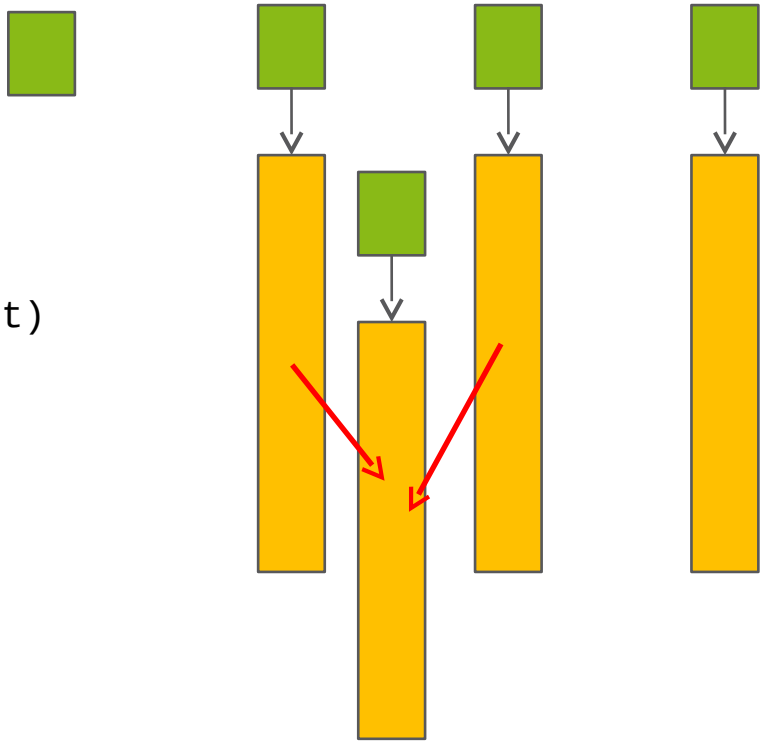
{
   Array res = left;
   res += right;
   return res;
}

void f()
{
   Array b, c, d;
   …
   Array a = b + c + d;
}
```



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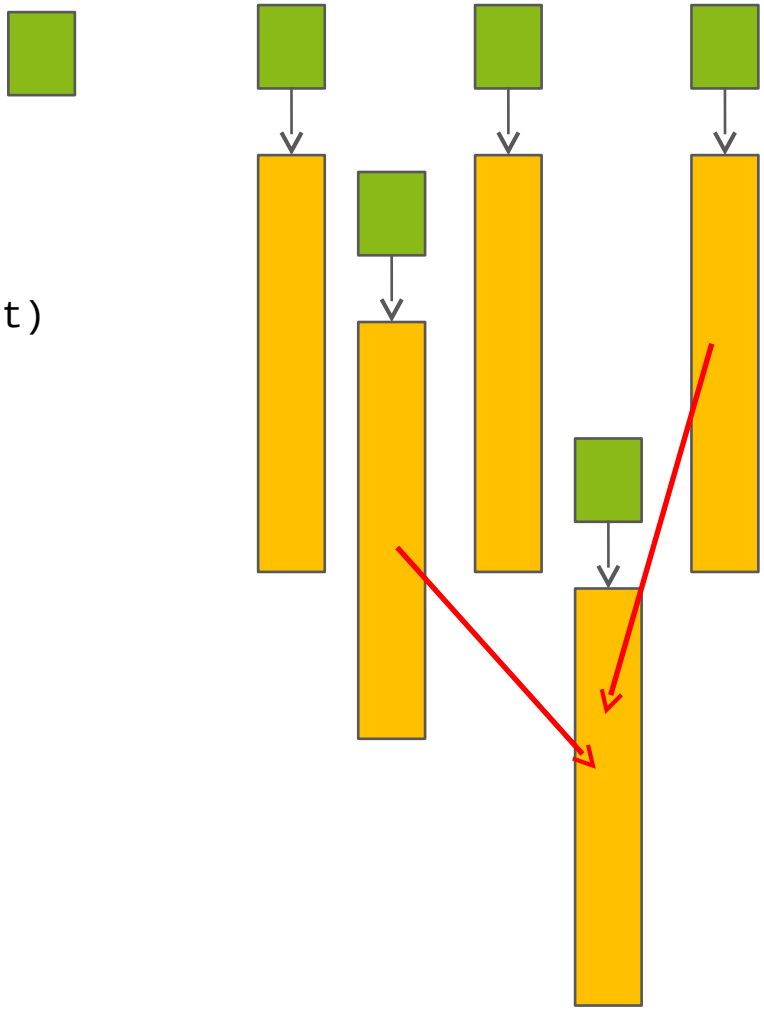
```
class Array
{
public:
   Array (const Array&);
   Array& operator=(const Array&);
   ~ Array ();
private:
   double  *val;
};
Array operator+(const Array& left, const Array& right)
{
   Array res = left;
   res += right;
   return res;
}

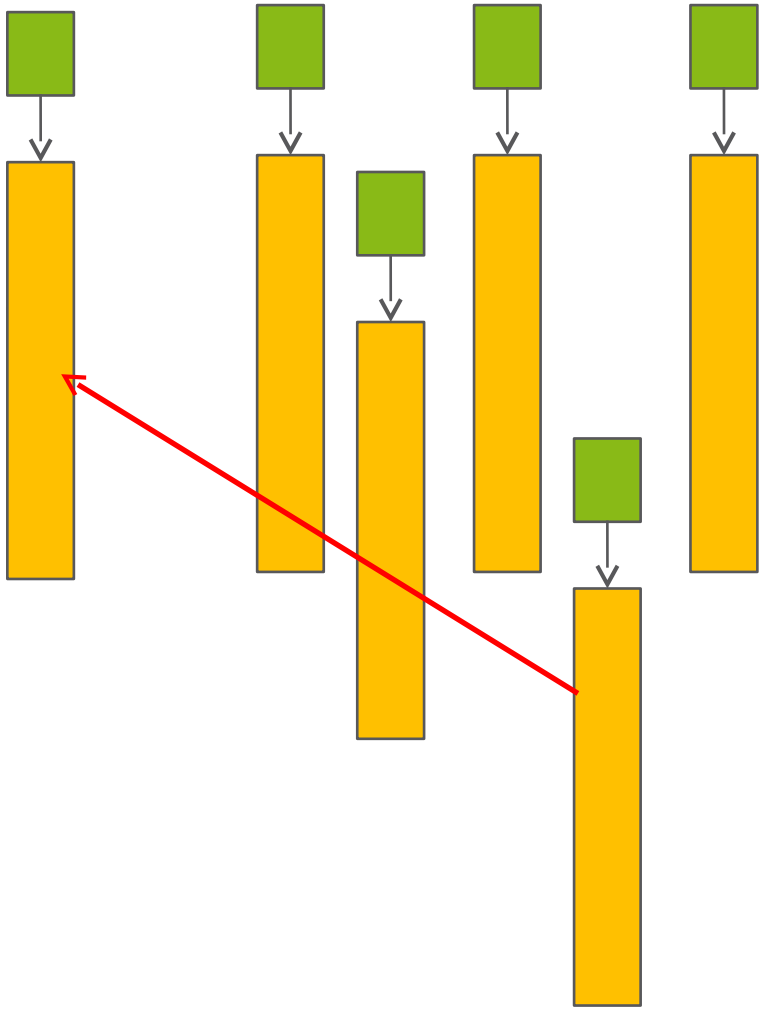
void f()
{
   Array b, c, d;
   …
   Array a = b + c + d;
}
```



# Value semantics

```
class Array
{
public:
   Array (const Array&);
   Array& operator=(const Array&);
   ~ Array ();
private:
   double  *val;
};
Array operator+(const Array& left, const Array& righ
{
   Array res = left;
   res += right;
   return res;
}

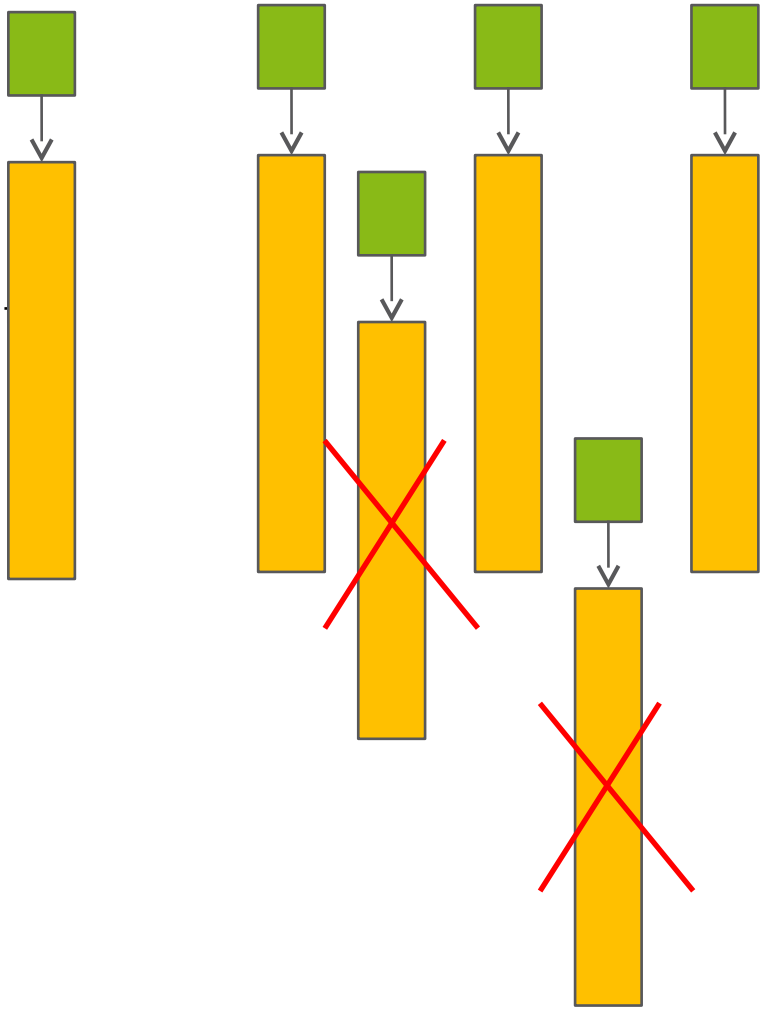
void f()
{
   Array b, c, d;
   …
   Array a = b + c + d;
}
```



# Value semantics

```
class Array
{
public:
   Array (const Array&);
   Array& operator=(const Array&);
   ~ Array ();
private:
   double  *val;
};
Array operator+(const Array& left, const Array& righ
{
   Array res = left;
   res += right;
   return res;
}

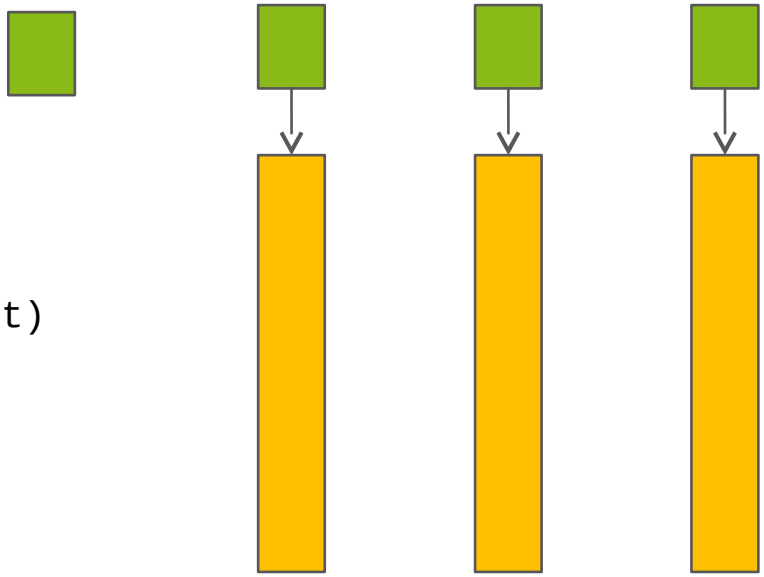
void f()
{
   Array b, c, d;
   …
   Array a = b + c + d;
}
```



# Move semantics

```
class Array
{
public:
   Array (const Array&);
   Array& operator=(const Array&);
   ~ Array ();
private:
   double  *val;
};
Array operator+(const Array& left, const Array& right)
{
   Array res = left;
   res += right;
   return res;
}

void f()
{
   Array b, c, d;
   …
   Array a = b + c + d;
}
```



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class Array
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public:
   Array (const Array&);
   Array& operator=(const Array&);
   ~ Array ();
private:
   double  *val;
};
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{
   Array res = left;
   res += right;
   return res;
}

void f()
{
   Array b, c, d;
   …
   Array a = b + c + d;
}
```
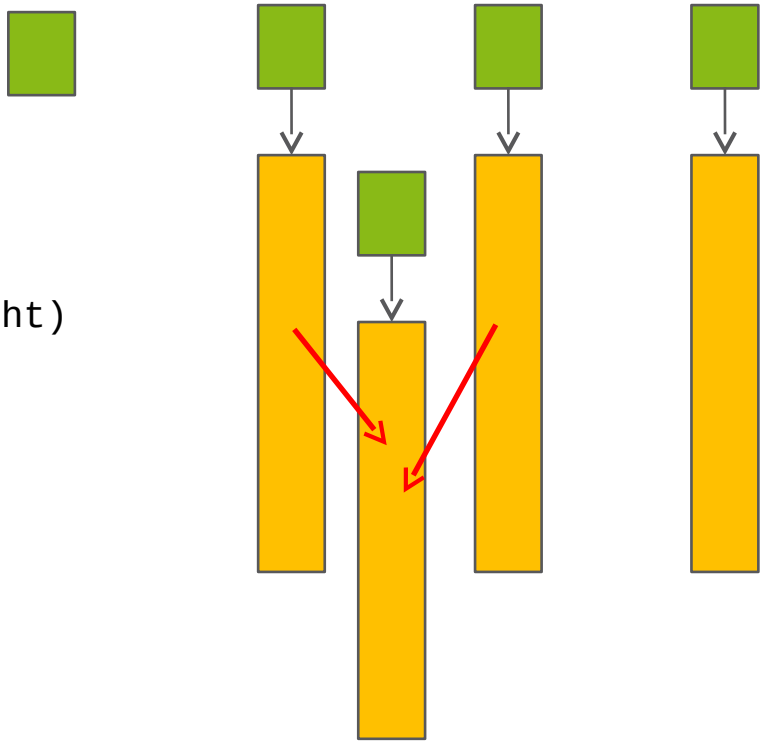
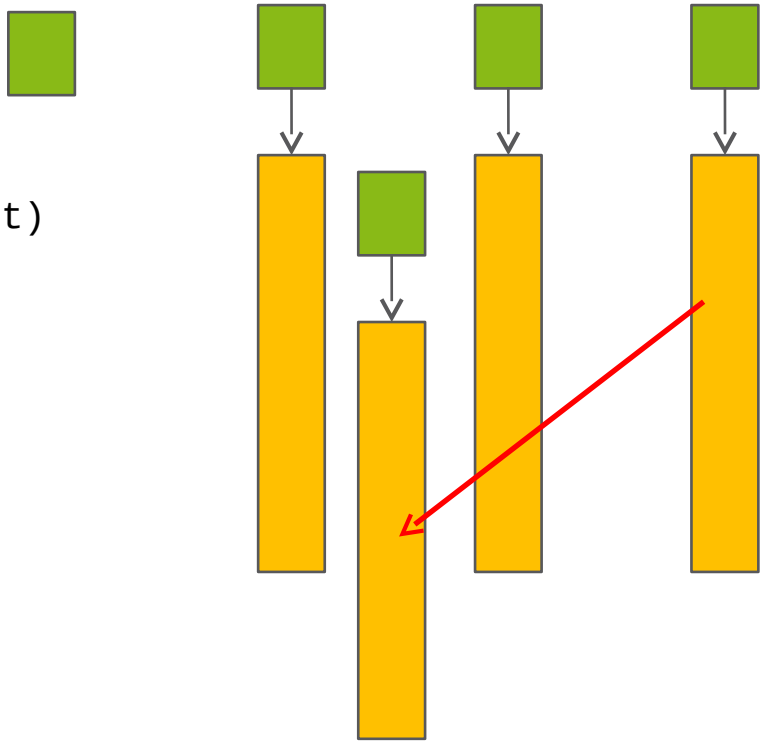



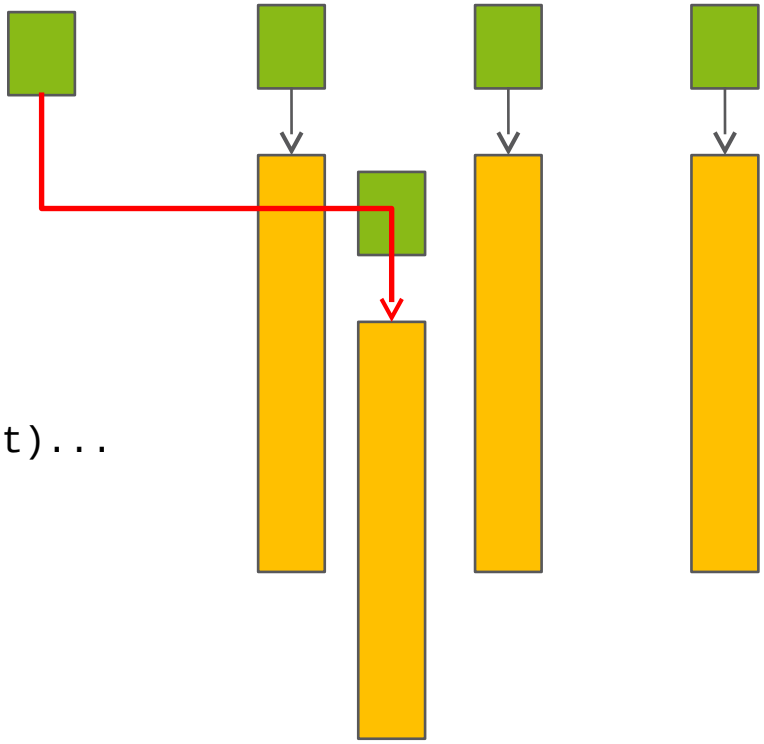
# Move semantics

```
class Array
{
public:
   Array (const Array&);
   Array& operator=(const Array&);
   …
};
Array operator+(const Array& left, const Array& right)
{
   Array res = left;
   res += right;
   return res;
}
Array operator+(Array&& left, const Array& right)
{
   left += right;
   return left;
}
void f()
{
   Array b, c, d;
   …
   Array a = b + c + d;
}
```



# Move semantics

```
class Array
{
public:
   Array (const Array&);
   Array (Array&&);
   Array& operator=(const Array&);
   Array& operator=(Array&&);
   ~ Array ();
private:
   double  *val;
};

Array operator+(const Array& left, const Array& right)...
Array operator+(Array&& left, const Array& right)...


void f()
{
   Array b, c, d;
   …
   Array a = b + c + d;
}
```



# Right value reference

- For overloading, we need a new type
  - Reference type for performance reasons
  - Overload resolution should prefer this new type on rvalue objects

```
void f(X& arg_)       // lvalue reference parameter
void f(X&& arg_)      // rvalue reference parameter
void f(const X& arg_) // const lvalue reference parameter

X x;
X g();

f(x);    // lvalue argument --> f(X&)
f(g());  // rvalue argument --> f(X&&)
```

# Move semantics

- Move semantics
  - Instead of copying steal the resources
  - Leave the other object in a destructible state
  - Rule of three becomes rule of five
  - All standard library components were extended

- Reverse compatibility
  - If we implement the old-style member functions with lvalue reference but do not implement the rvalue reference overloading versions
    we keep the old behaviour -> gradually move to move semantics.
  - If we implement only rvalue operations we cannot call these on lvalues -> no default copy ctor or **operator**= will be generated.

- Serious performance gain
  - Except some rare RVO situations

# Special member functions

```
class X
{
public:
  X(const X& rhs);
  X(X&& rhs);

  X& operator=(const X& rhs);    // = default  or = delete
  X& operator=(X&& rhs);
private:
  // ...
};
X& X::operator=(const X& rhs)
{
  // free old resources than allocate and copy resource from rhs
  return *this;
}
X& X::operator=(X&& rhs)  // draft version, will be revised
{
  // free old resources than move resource from rhs
  // leave rhs in a valid, destructable state
  return *this;
}
```

# Generation of special member functions

1. The two **copy operations** (copy constructor and copy assignment) **are independent.** Declaring copy constructor does nor prevent compiler to generate copy assignment (and vice versa). (same as in C++98)

2. **Move operations are not independent**. Declare either prevents the compiler to generate the other.

3. If any of the **copy operation is declared**, then **none of the move** operation will be generated.

4. If any of the **move operation is declared**, then **none of the copy** operation will be generated. This is the opposite rule of (3).

5. If a **destructor is declared**, than **none of the move** operation will be generated. Copy operations are still generated for reverse compatibility with C++98.

6. **Default constructor** generated only no constructor is declared (same as in C++98)

# Move operations

- For reverse compatibility, move operations are generated only when
  - No copy operations are declared
  - No move operations are declared
  - No destructor is declared
- Function templates  do not considered here
  - Templated copy constructor, assignment does not prevent move operation generations
  - Same rule since C++98 with copy operations
- Play safe!

  … it is easy...
- Really?

# A simple program

```
#include <iostream>
#include <vector>

struct S
{
    S() { a = ++cnt; }
    int a;
    static int cnt;
};
int S::cnt = 0;

int main()
{
    std::vector<S>  sv(5);
    sv.push_back(S());

    for (std::size_t i = 0; i< sv.size(); ++i)
        std::cout << sv[i].a << " ";
    std::cout << std::endl;
}
```

# First amendment to the C++ standard

"The committee shall make no rule that prevents C++ programmers from shooting themselves in the foot."

quoted by Thomas Becker

http://thbecker.net/articles/rvalue_references/section_04.html

# std::move

```
struct S
{
    S() { a = ++cnt; std::cout << "S() ";  }
    S(const S& rhs)    { a = rhs.a; std::cout << "copyCtr "; }
    S(S&& rhs)         { a = rhs.a; std::cout << "moveCtr "; }
    S& operator=(const S& rhs) { a = rhs.a; std::cout << "copy= "; return *this; }
    S& operator=(S&& rhs)      { a = rhs.a; std::cout << "move= "; return *this; }
    int a;
    static int cnt;
};
int S::cnt = 0;

template<class T>
void swap(T& a, T& b)
{
    T tmp(a);
    a = b;
    b = tmp;
}
int main()
{
    S a, b;
    swap( a, b);
}
```

# std::move

```
struct S
{
    S() { a = ++cnt; std::cout << "S() ";  }
    S(const S& rhs)    { a = rhs.a; std::cout << "copyCtr "; }
    S(S&& rhs)         { a = rhs.a; std::cout << "moveCtr "; }
    S& operator=(const S& rhs) { a = rhs.a; std::cout << "copy= "; return *this; }
    S& operator=(S&& rhs)      { a = rhs.a; std::cout << "move= "; return *this; }
    int a;
    static int cnt;
};
int S::cnt = 0;

template<class T>
void swap(T& a, T& b)
{
    T tmp(a);
    a = b;
    b = tmp;
}
int main()
{
    S a, b;
    swap( a, b);
}
```

```
$ ./a.out
S() S() copyCtr copy= copy=
```

# std::move

```
struct S
{
    S() { a = ++cnt; std::cout << "S() ";  }
    S(const S& rhs)    { a = rhs.a; std::cout << "copyCtr "; }
    S(S&& rhs)         { a = rhs.a; std::cout << "moveCtr "; }
    S& operator=(const S& rhs) { a = rhs.a; std::cout << "copy= "; return *this; }
    S& operator=(S&& rhs)      { a = rhs.a; std::cout << "move= "; return *this; }
    int a;
    static int cnt;
};
int S::cnt = 0;

template<class T>
void swap(T& a, T& b)
{
    T tmp(a);
    a = b;
    b = tmp;
}
int main()
{
    S a, b;
    swap( a, b);
}
```

```
$ ./a.out
S() S() copyCtr copy= copy=
```

If it has a name: LVALUE

# std::move

```
struct S
{
    S() { a = ++cnt; std::cout << "S() ";  }
    S(const S& rhs)    { a = rhs.a; std::cout << "copyCtr "; }
    S(S&& rhs)         { a = rhs.a; std::cout << "moveCtr "; }
    S& operator=(const S& rhs) { a = rhs.a; std::cout << "copy= "; return *this; }
    S& operator=(S&& rhs)      { a = rhs.a; std::cout << "move= "; return *this; }
    int a;
    static int cnt;
};
int S::cnt = 0;

template<class T>
void swap(T& a, T& b)
{
    T tmp(std::move(a));
    a = std::move(b);
    b = std::move(tmp);
}
int main()
{
    S a, b;
    swap( a, b);
}
```

# std::move

```
struct S
{
    S() { a = ++cnt; std::cout << "S() ";  }
    S(const S& rhs)    { a = rhs.a; std::cout << "copyCtr "; }
    S(S&& rhs)         { a = rhs.a; std::cout << "moveCtr "; }
    S& operator=(const S& rhs) { a = rhs.a; std::cout << "copy= "; return *this; }
    S& operator=(S&& rhs)      { a = rhs.a; std::cout << "move= "; return *this; }
    int a;
    static int cnt;
};
int S::cnt = 0;

template<class T>
void swap(T& a, T& b)
{
    T tmp(std::move(a));
    a = std::move(b);
    b = std::move(tmp);
}
int main()
{
    S a, b;
    swap( a, b);
}
```

```
$ ./a.out
S() S() moveCtr move= move=
```

# std::move(x)

- Right value reference cast
- Usually has positive effect of performance
  - Many standard lib function utilize right-value references
- Sometimes we have to use it
  - Movable non-copyable classes
  - std::unique_ptr, std::fstream, std::thread
- Might be dangerous
  - A variable with name left with unspecified value

# Rvalue and constness

```
#include <iostream>

struct MyString {
    MyString() { std::cerr << "defCtor" << std::endl; }
    MyString(const MyString& rhs) { std::cerr << "copyCtor" << std::endl; }
    MyString(MyString&& rhs) { std::cerr << "moveCtor" << std::endl; }
};
class MoveConstr {
public:
    MoveConstr( MyString s) : value(std::move(s)) { }
private:
    MyString value;
};
int main() {
    MoveConstr mc(MyString());
    return 0;
}
```

# Most vexing parse

```
MoveConstr mc(MyString())
```

- Function declaration:
  - mc returning MoveConstr type
  - One parameter: a pointer to function returns MyString and has no parameter

```
$ clang++ -std=c++11 -Wvexing-parse m1.cpp
m1.cpp:21:16: warning: parentheses were disambiguated as a function declaration
      [-Wvexing-parse]
  MoveConstr mc(MyString());
               ^~~~~~~~~~~~
m1.cpp:21:17: note: add a pair of parentheses to declare a variable
  MoveConstr mc(MyString());
                ^
                (         )
1 warning generated.
```

# Rvalue and constness

```
#include <iostream>

struct MyString {
    MyString() { std::cerr << "defCtor" << std::endl; }
    MyString(const MyString& rhs) { std::cerr << "copyCtor" << std::endl; }
    MyString(MyString&& rhs) { std::cerr << "moveCtor" << std::endl; }
};
class MoveConstr {
public:
    MoveConstr( MyString s) : value(std::move(s)) { }
private:
    MyString value;
};
int main() {
    MoveConstr mc{ MyString() };     // MoveConstr mc((MyString()));
    return 0;
}
$ ./a.out
defCtor
moveCtor
```

# Rvalue and constness

```
#include <iostream>

struct MyString {
    MyString() { std::cerr << "defCtor" << std::endl; }
    MyString(const MyString& rhs) { std::cerr << "copyCtor" << std::endl; }
    MyString(MyString&& rhs) { std::cerr << "moveCtor" << std::endl; }
};
class MoveConstr {
public:
    MoveConstr( const MyString s) : value(std::move(s)) { }
private:
    MyString value;
};
int main() {
    MoveConstr mc{ MyString() };
    return 0;
}
$ ./a.out
defCtor
copyCtor
```

# Inheritance

```
class Base
{
public:
    Base(const Base& rhs);    // non-move semantics
    Base(Base&& rhs);            // move semantics
};
class Derived : public Base
{
    Derived(const Derived& rhs);  // non-move semantics
    Derived(Derived&& rhs);          // move semantics
};
Derived(Derived const & rhs) : Base(rhs)  // non-move semantics
{
// copy derived specific…
}
Derived(Derived&& rhs) : Base(rhs)           // move semantics
{
// move derived specific…
}
```

# Inheritance

```
class Base
{
public:
    Base(const Base& rhs);    // non-move semantics
    Base(Base&& rhs);            // move semantics
};
class Derived : public Base
{
    Derived(const Derived& rhs);  // non-move semantics
    Derived(Derived&& rhs);          // move semantics
};
Derived(Derived const & rhs) : Base(rhs)  // non-move semantics
{
// copy derived specific…
}
Derived(Derived&& rhs) : Base(rhs)           // wrong!
{
// move derived specific…
}
```

# Inheritance

```
class Base
{
public:
    Base(const Base& rhs);    // non-move semantics
    Base(Base&& rhs);            // move semantics
};
class Derived : public Base
{
    Derived(const Derived& rhs);  // non-move semantics
    Derived(Derived&& rhs);          // move semantics
};
Derived(Derived const & rhs) : Base(rhs)  // non-move semantics
{
// copy derived specific…
}
Derived(Derived&& rhs) : Base(std::move(rhs))        // good, calls Base(Base&& rhs)
{
// move derived specific…
}
```

# Vector

```
#include <iostream>
#include <vector>

struct S
{
    S() { a = ++cnt; std::cout << "S() ";  }
    S(const S& rhs)    { a = rhs.a; std::cout << "copyCtr ";  }
    S(S&& rhs)         { a = rhs.a; std::cout << "moveCtr "; }
    S& operator=(const S& rhs) { a = rhs.a; std::cout << "copy= "; return *this; }
    S& operator=(S&& rhs)      { a = rhs.a; std::cout << "move= "; return *this; }
    int a;
    static int cnt;
};
int S::cnt = 0;

int main()
{
    std::vector<S>  sv(5);
    sv.push_back(S());

    for (std::size_t i = 0; i < sv.size(); ++i)
        std::cout << sv[i].a << " ";
    std::cout << std::endl;
}
```

# Vector

```
#include <iostream>
#include <vector>

struct S
{
    S() { a = ++cnt; std::cout << "S() ";  }
    S(const S& rhs)    { a = rhs.a; std::cout << "copyCtr ";  }
    S(S&& rhs)         { a = rhs.a; std::cout << "moveCtr "; }
    S& operator=(const S& rhs) { a = rhs.a; std::cout << "copy= "; return *this; }
    S& operator=(S&& rhs)      { a = rhs.a; std::cout << "move= "; return *this; }
    int a;
    static int cnt;
};
int S::cnt = 0;

int main()
{
    std::vector<S>  sv(5);
    sv.push_back(S());

    for (std::size_t i = 0; i < sv.size(); ++i)
        std::cout << sv[i].a << " ";
    std::cout << std::endl;
}
```

```
$ g++ -std=c++11  vec.cpp &&  ./a.out
S() S() S() S() S() S() moveCtr
copyCtr copyCtr copyCtr copyCtr copyCtr
1 2 3 4 5 6
```

# Vector

```
#include <iostream>
#include <vector>

struct S
{
    S() { a = ++cnt; std::cout << "S() ";  }
    S(const S& rhs)    { a = rhs.a; std::cout << "copyCtr ";  }
    S(S&& rhs)         { a = rhs.a; std::cout << "moveCtr "; }
    S& operator=(const S& rhs) { a = rhs.a; std::cout << "copy= "; return *this; }
    S& operator=(S&& rhs)      { a = rhs.a; std::cout << "move= "; return *this; }
    int a;
    static int cnt;
};
int S::cnt = 0;

int main()
{
    std::vector<S>  sv(5);
    sv push_back(S());

    for (std::size_t i = 0; i < sv.size(); ++i)
        std::cout << sv[i].a << " ";
    std::cout << std::endl;
}
```

```
$ g++ -std=c++11  vec.cpp &&  ./a.out
S() S() S() S() S() S() moveCtr
copyCtr copyCtr copyCtr copyCtr copyCtr
1 2 3 4 5 6
```

# Vector

```
#include <iostream>
#include <vector>

struct S
{
    S() { a = ++cnt; std::cout << "S() ";  }
    S(const S& rhs)    { a = rhs.a; std::cout << "copyCtr ";  }
    S(S&& rhs)   noexcept    { a = rhs.a; std::cout << "moveCtr "; }
    S& operator=(const S& rhs) { a = rhs.a; std::cout << "copy= "; return *this; }
    S& operator=(S&& rhs)      { a = rhs.a; std::cout << "move= "; return *this; }
    int a;
    static int cnt;
};
int S::cnt = 0;

int main()
{
    std::vector<S>  sv(5);
    sv push_back(S());

    for (std::size_t i = 0; i < sv.size(); ++i)
        std::cout << sv[i].a << " ";
    std::cout << std::endl;
}
```

# Vector

```
#include <iostream>
#include <vector>

struct S
{
    S() { a = ++cnt; std::cout << "S() ";  }
    S(const S& rhs)    { a = rhs.a; std::cout << "copyCtr ";  }
    S(S&& rhs)   noexcept    { a = rhs.a; std::cout << "moveCtr "; }
    S& operator=(const S& rhs) { a = rhs.a; std::cout << "copy= "; return *this; }
    S& operator=(S&& rhs)      { a = rhs.a; std::cout << "move= "; return *this; }
    int a;
    static int cnt;
};
int S::cnt = 0;

int main()
{
    std::vector<S>  sv(5);
    sv.push_back(S());

    for (std::size_t i = 0; i < sv.size(); ++i)
        std::cout << sv[i].a << " ";
    std::cout << std::endl;
}
```

```
$ g++ -std=c++11  vec.cpp &&  ./a.out
S() S() S() S() S() S()
moveCtr moveCtr moveCtr moveCtr moveCtr moveCtr
1 2 3 4 5 6
```

# std::move(b,e,b2)

```
#include <iostream>
#include <vector>
#include <list>
#include <algorithm>

struct S
{
    S() { a = ++cnt; std::cout << "S() ";  }
    S(const S& rhs) { a = rhs.a; std::cout << "copyCtr ";  }
    S(S&& rhs)   noexcept    { a = rhs.a; std::cout << "moveCtr "; }
    S& operator=(const S& rhs) { a = rhs.a; std::cout << "copy= "; return *this; }
    S& operator=(S&& rhs)      { a = rhs.a; std::cout << "move= "; return *this; }
    int a ;
    static int cnt;
};
int S::cnt = 0;

int main()
{
    std::list<S> sl = { S(), S(), S(), S(), S() };
    std::vector<S> sv(5);
    std::move( sl.begin(), sl.end(), sv.begin());

    for (const S& s : sv)
        std::cout << s.a << " ";
     std::cout << std::endl;
}
```

# std::move(b,e,b2)

```
#include <iostream>
#include <vector>
#include <list>
#include <algorithm>

struct S
{
    S() { a = ++cnt; std::cout << "S() ";  }
    S(const S& rhs) { a = rhs.a; std::cout << "copyCtr ";  }
    S(S&& rhs)   noexcept    { a = rhs.a; std::cout << "moveCtr "; }
    S& operator=(const S& rhs) { a = rhs.a; std::cout << "copy= "; return *this; }
    S& operator=(S&& rhs)      { a = rhs.a; std::cout << "move= "; return *this; }
    int a ;
    static int cnt;
};
int S::cnt = 0;

int main()
{
    std::list<S> sl = { S(), S(), S(), S(), S() };
    std::vector<S> sv(5);
    std::move( sl.begin(), sl.end(), sv.begin());

    for (const S& s : sv)
        std::cout << s.a << " ";
     std::cout << std::endl;
}
```

```
$ g++ -std=c++11  &&  ./a.out
S() S() S() S() S() copyCtr
copyCtr copyCtr copyCtr copyCtr
S() S() S() S() S()
move= move= move= move= move=
1 2 3 4 5
```

# std::move(b,e,b2)

```
#include <iostream>
#include <vector>
#include <set>
#include <algorithm>

struct S
{
    S() { a = ++cnt; std::cout << "S() ";  }
    S(const S& rhs) { a = rhs.a; std::cout << "copyCtr ";  }
    S(S&& rhs)   noexcept    { a = rhs.a; std::cout << "moveCtr "; }
    S& operator=(const S& rhs) { a = rhs.a; std::cout << "copy= "; return *this; }
    S& operator=(S&& rhs)      { a = rhs.a; std::cout << "move= "; return *this; }
    int a ;
    static int cnt;
};
int S::cnt = 0;
bool operator<(const S& x, const S& y) { return x.a < y.a; }
int main()
{
    std::set<S> sl = { S(), S(), S(), S(), S() };
    std::vector<S> sv(5);
    std::move( sl.begin(), sl.end(), sv.begin());

    for (const S& s : sv)
        std::cout << s.a << " ";
     std::cout << std::endl;
}
```

# std::move(b,e,b2)

```
#include <iostream>
#include <vector>
#include <set>
#include <algorithm>

struct S
{
    S() { a = ++cnt; std::cout << "S() ";  }
    S(const S& rhs) { a = rhs.a; std::cout << "copyCtr ";  }
    S(S&& rhs)   noexcept    { a = rhs.a; std::cout << "moveCtr "; }
    S& operator=(const S& rhs) { a = rhs.a; std::cout << "copy= "; return *this; }
    S& operator=(S&& rhs)      { a = rhs.a; std::cout << "move= "; return *this; }
    int a ;
    static int cnt;
};
int S::cnt = 0;
bool operator<(const S& x, const S& y) { return x.a < y.a; }
int main()
{
    std::set<S> sl = { S(), S(), S(), S(), S() };
    std::vector<S> sv(5);
    std::move( sl.begin(), sl.end(), sv.begin());

    for (const S& s : sv)
        std::cout << s.a << " ";
     std::cout << std::endl;
}
```

```
$ g++ -std=c++11  &&  ./a.out
S() S() S() S() S() copyCtr
copyCtr copyCtr copyCtr copyCtr
S() S() S() S() S()
copy= copy= copy= copy= copy=
1 2 3 4 5
```

# Overloading on right value

```
T &      value(optional<T> &      par_);
T &&     value(optional<T> &&     par_);
T const& value(optional<T> const& par_);
```

But in object-oriented programming, sometimes we want to overload on the **this** parameter too.

```
template <typename T>
class optional
{
  // ...
  T&       value() &;
  T&&      value() &&;
  T const& value() const&;
};
```

# RVO

```
std::vector<double>  fill();

int main()
{
    std::vector<double> vd = fill();
    // ...
}

std::vector<double>  fill()
{
   std::vector<double> local;
   // fill the elements
   return local;      // return or move?
}
```

David Abrahams wrote an article on this:
https://www.scribd.com/document/316704010/Want-Speed-Pass-by-Value
(originally at: http://cpp-next.com/archive/2009/08/want-speed-pass-by-value/)

# Forwarding/Universal reference

Forwarding/Universal reference is used in case of type deduction

```
class X;
Void f(X&& param)        // rvalue reference
X&& var1 = X();          // rvalue reference
auto&& var2 = var1;      // NOT rvalue reference: universal reference
template <typename T>
void f(std::vector<T>&& param); // rvalue reference (1)
template <typename T>
void f(T&& param);        // NOT rvalue reference: universal reference (2)
template <typename T>
void f(const T&& param); // rvalue reference

X var;
f(var);                  // lvalue passed: param type is: X& (2)
f(std::move(var));       // rvalue passed: param type is: X&& (2)
std::vector<int> v;
f(v);                    // syntax error: can't bind lvalue to rvalue  (1)
```

# Universal reference

Universal reference is used in case of type deduction

```
template <class T, class Allocator = allocator<T>>
class vector
{
public:
  void push_back(T&& x);  // rvalue reference, no type deduction here
  // ...
};

template <class T, class Allocator = allocator<T>>
class vector
{
public:
  template <class... Args>
  void emplace_back(Args&&... args);    // universal reference, type deduction
  // ...
};
```

# Perfect forwarding

```
template<typename T, typename Arg>
shared_ptr<T> factory(Arg arg)
{
  return shared_ptr<T>(new T(arg));    // call T(arg) by value. Bad!
}

// A half-good solution is passing arg by reference:
template<typename T, typename Arg>
shared_ptr<T> factory(Arg& arg)
{
  return shared_ptr<T>(new T(arg));
}

But this does not work for rvalue parameters:

factory<X>(f());        // error if f() returns by value
factory<X>(42);         // error
```

# Perfect forwarding

```
// If f() called on "lvalue of A"  T --> A&     argument type --> A&
// If f() called on "rvalue of A"  T --> A      argument type --> A

template<typename T, typename Arg>
shared_ptr<T> factory(Arg&& arg)
{
  return shared_ptr<T>(new T(std::forward<Arg>(arg)));
}
template<class S>
S&& forward(typename remove_reference<S>::type& a) noexcept
{
  return static_cast<S&&>(a);
}

// Reference collapsing:

A& &   --> A&
A& &&  --> A&
A&& &  --> A&
A&& && --> A&&
```

# Perfect forwarding

```
shared_ptr<A> factory(X&& arg)
{
  return shared_ptr<A>(new A(std::forward<X>(arg)));
}
X&& forward(X& a) noexcept  // std::forward keeps move semantic
{
  return static_cast<X&&>(a);
}

template <typename T>           // C++11
typename remove_reference<T>::type&&
std::move(T&& a) noexcept
{
  typedef typename remove_reference<T>::type&& RvalRef;
  return static_cast<RvalRef>(a);
}
template <typename T>           // C++14
decltype(auto) std::move(T&& a)
{
  using ReturnType = remove_reference_t<T>&&;
  return static_cast<ReturnType>(a);
}
```